3.22 Integral Ring Extensions

Definition 3.22.1 (Integral Element)
*Let $\phi : A \to B$ be a ring map and $\alpha \in B$. Then we say α is **algebraic** over A if $m(\alpha) = 0$ for some polynomial $m(X) \in A[X]$.*

*Furthermore we say that α is **integral** over A if $m(X)$ may be chosen to be monic.*

Note often we assume $A \subseteq B$ and ϕ is the identity.

Definition 3.22.2 (Ring Extensions)
Let $\phi : A \to B$ be a ring map (so that B is an A-algebra). Then we say ϕ is

- ***finite** if B is finite as an A-module*
- ***finite-type** if B is finitely generated as an A-algebra*
- ***integral** if every element of B is integral over A*

When $A \subseteq B$ and ϕ is the identity then we say that B is respectively finite over A, finite-type over A or integral over A.

We note the trivial implication

$$\text{finite} \implies \text{finite type}$$

For example $k[X]$ is a ring of finite type over k, but certainly not finite. The following criterion for integrality is fundamental.

Proposition 3.22.3
Let $\phi : A \to B$ be a ring map and $b \in B$. Then the following are equivalent

a) b is integral over A
b) $\phi(A)[b]$ is a finite A-module
c) $\phi(A)[b]$ is contained in a subring C of B which is a finite A-module
d) There exists a $\phi(A)[b]$-module M which is faithful and finite as an A-module

Proof. Note that the subring C in c) is a faithful A-module, so the only non-trivial step is d) $\implies$ a). This is the usual "determinant trick". We apply (3.16.4) by considering $\psi_b \in \text{End}_A(M)$ to be multiplication by b. Then we have some monic polynomial $P(X) \in A[X]$ such that $P(\psi_b) = 0$, whence $P^\phi(b)m = 0$ for all $m \in M$. Since M is faithful, then we have $P(b) = 0$ as required. ∎

Proposition 3.22.4 (Finite $\iff$ finite-type and integral)
Let $\phi : A \to B$ be a ring map and $b_1, ..., b_n \in B$ integral over A. Then the ring homomorphism $\phi : A \to \phi(A)[b_1, ..., b_n]$ is finite and integral.

In particular if ϕ is integral and of finite type if and only if it is finite.

Proof. We assume without loss of generality that $A \subseteq B$ and ϕ is the identity map. Consider a tower

$$A \subset A[b_1] \subset ... \subset A[b_1, ..., b_n] = B$$

We proceed inductively on n. Namely we assume that $A[b_1, ..., b_i]$ is a finite A-module. Then a-fortiori $A[b_1, ..., b_{i+1}]$ is integral over $A[b_1, ..., b_i]$. Therefore by the previous Proposition it is a finite $A[b_1, ..., b_i]$-module and therefore a finite A-module (by (3.13.5)).

For any $b \in A[b_1, ..., b_n]$ we have $A[b] \subset A[b_1, ..., b_n]$ so by (3.22.3) we have b is integral over A.

It then follows that B integral and finite type $\implies$ B is finite (as an A-module). Conversely if B is finite then it is clearly finitely-generated. Further for any $b \in B$ then $A[b]$ is contained in the ring B which is finite as an A-module, and therefore is b is integral by (3.22.3). ∎

Proposition 3.22.5 (Transitivity property)
Let $\phi : A \to B$ and $\psi : B \to C$ be ring maps.

If $c \in C$ is integral over B, then it is integral over A (with respect to the ring map $\psi \circ \phi : A \to C$)

In particular if ϕ integral and ψ integral (e.g. surjective) $\implies$ $\psi \circ \phi$ integral.

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Proof. Suppose $c \in C$ is integral over B. Let $b_0, ..., b_{n-1}$ be the coefficients of the integral relation then clearly c is integral over $B' \coloneqq A[b_0, ..., b_{n-1}]$. By (3.22.3) B' is a finite A-module and $B'[c]$ is a finite B'-module. Therefore $B'[c]$ is a finite A-module and by (3.22.3) we have c is integral over A. ∎

Definition 3.22.6 (Integrally Closed)
*Let $A \subset B$ be a subring, then we say that A is **integrally closed** in B if*

$$b \in B \text{ integral over } A \implies b \in A.$$

*We say that an integral domain A is **integrally closed** if it is integrally closed in its quotient field.*

Proposition 3.22.7 (Integral Closure)
*Let A be a subring of a ring B. Then the set of elements of B integral over A (the **integral closure**) is a subring of B. Denote this by $\overline{A}^B$ or just $\overline{A}$ when the overring B is unambiguous.*

*Further $\overline{A}^B$ is **integrally closed** in B.*

Proof. Let $\alpha, \beta \in B$ be integral over A. Then by (3.22.4) the subring $A[\alpha, \beta]$ is a finite A-module containing $\alpha \pm \beta$ and $\alpha\beta$. Therefore by (3.22.3) they are also integral over A.

Clearly $\overline{A}$ is integral over A so by (3.22.5) it is integrally closed. ∎

Lemma 3.22.8
Let A be a subring of a field K and $S \subset A$ a multiplicatively closed subset. If $x \in K$ is integral over $S^{-1}A$ then sx is integral over A for some $s \in S$.

Conversely if $y \in K$ is integral over A then $\frac{y}{s}$ is integral over $S^{-1}A$ for every $s \in S$.

Proof. Suppose $x \in K$ is integral over $S^{-1}A$ then we have by definition

$$x^n + \sum_{j=0}^{n-1} \frac{a_j}{s_j} x^j = 0$$

for $a_j \in A$ and $s_j \in S$. Let $s \coloneqq s_0 ... s_{n-1}$ and multiply by s^n to find

$$y^n + \sum_{j=0}^{n-1} s^{n-j-1} \frac{s}{s_j} a_j y^j = 0$$

where $y = sx$. This shows that sx is integral over A

Similarly if $y \in K$ is integral over A then

$$y^n + \sum_{j=0}^{n-1} a_j y^j = 0$$

for some $a_j \in A$. Then let $x = \frac{y}{s}$ we have

$$x^n + \sum_{j=0}^{n-1} \frac{a_j}{s^{n-j}} x^j = 0$$

whence x is integral over $S^{-1}A$. ∎

Lemma 3.22.9
Let $A \subset K$ be a subring of a field. Then $\overline{S^{-1}A}^K = S^{-1}\overline{A}^K$.

In particular if A is integrally closed in K then so is $S^{-1}A$.

Proof. This follows directly from (3.22.8). ∎

Proposition 3.22.10 (Normality is Local)
Let A be an integral domain. Then the following are equivalent

a) A is integrally closed
b) $A_{\mathfrak{p}}$ is integrally closed for every prime ideal $\mathfrak{p}$
c) $A_{\mathfrak{m}}$ is integrally closed for every maximal ideal $\mathfrak{m}$

Proof. a) $\implies$ b) This is simply (3.22.9).

b) $\implies$ c) is immediate.

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c) $\implies$ a) Suppose that $x \in K$ is integral over A and consider the ideal $\mathfrak{a} := \{a \in A \mid ax \in A\}$. Then by definition it is a non-zero ideal, and we wish to show that $\mathfrak{a} = A$. Suppose not, then by (...) it is contained in some maximal ideal $\mathfrak{m}$. A-fortiori x is integral over $A_{\mathfrak{m}}$ whence by assumption $x \in A_{\mathfrak{m}}$. This means there is some $s \notin \mathfrak{m}$ such that $sx \in A$, that is $s \in \mathfrak{a}$ a contradiction. ∎

Proposition 3.22.11 (Integral closure of an ideal)
Let $A \subset B$ be a subring and $\mathfrak{a} \triangleleft A$ an ideal. Then for $b \in B$ TFAE

a) b integral over $\mathfrak{a}$
b) b^n integral over $\mathfrak{a}$ for some $n \geq 1$
c) $b \in \sqrt{\mathfrak{a}\overline{A}}$

In particular $\overline{\mathfrak{a}}$ is an ideal of $\overline{A}$.

Furthermore if A is integrally closed in B then $\overline{\mathfrak{a}} = \sqrt{\mathfrak{a}}$.

Proof. It's clear that a) $\iff$ b). For a) $\implies$ c) consider the integral relation

$$b^n + a_{n-1}b^{n-1} + ...a_0 = 0 \quad a_i \in \mathfrak{a}$$

By (3.22.7) $\overline{A}$ is a subring, and by assumption $b \in \overline{A}$. Therefore $b^k \in \overline{A}$, and the integral relation shows that $b^n \in \mathfrak{a}\overline{A}$ as required.

For c) $\implies$ b) suppose $b \in \sqrt{\mathfrak{a}\overline{A}}$ then $b^n = \sum_{i=1}^n a_i x_i$ for $a_i \in \mathfrak{a}$ and $x_i \in \overline{A}$. Let $B' := A[x_1, ..., x_n]$, which is a finite A-submodule by (3.22.4). Let $\phi \in \mathrm{End}_A(B')$ denote multiplication by b^n then $\phi(B') \subseteq \mathfrak{a}B'$ so by (3.16.4) ϕ satisfies a monic polynomial with coefficients in $\mathfrak{a}$. In particular b^n is integral over $\mathfrak{a}$. ∎

Proposition 3.22.12
A UFD is integrally closed.

In particular polynomial ring over a UFD is integrally closed.

Proof. **TODO** ∎

The following criterion is also useful:

Lemma 3.22.13 (Integral Criterion II)
Let $\phi : A \to B$ be a ring map. Suppose $x \in B$ is invertible, then x is integral over A if and only if $x \in \phi(A)[x^{-1}]$

Proof. Suppose $x \in \phi(A)[x^{-1}]$ then

$$x = \phi(a_0) + \phi(a_1)x^{-1} + ... + \phi(a_n)x^{-n}$$

Multiply by x^n to deduce an integral equation. Conversely suppose $x \in B$ is integral over A then by definition

$$x^n + \phi(a_{n-1})x^{n-1} + ... + \phi(a_0) = 0$$

Multiply by $x^{-(n-1)}$ to deduce $x \in \phi(A)[x^{-1}]$ ∎

Proposition 3.22.14 (Integral extension preserves field property)
Let $\phi : A \hookrightarrow B$ be an injective, integral ring map. Then B is a field if and only if A is a field.

Proof. As ϕ is injective, it induces an isomorphism between A and $\phi(A)$. So we may assume without loss of generality that $A \subseteq B$ and ϕ is the identity.

Suppose B is a field and $x \in A$. Then $x^{-1} \in B$ is integral over A by hypothesis, so by the previous Lemma $x^{-1} \in A[x] \subseteq A$. Therefore A is a field.

Conversely suppose A is a field and $0 \neq x \in B$. Then by hypothesis x is integral over A, that is

$$x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0 = 0$$

Choose the degree n to be minimal. We claim $a_0 \neq 0$, for if $a_0 = 0$ we may cancel x to obtain an integral relation of smaller degree. Therefore

$$-x(x^{n-1} + a_{n-1}x^{n-2} + ... + a_1)a_0^{-1} = 1$$

and in particular x is invertible. ∎

Proposition 3.22.15
Let $\phi : A \to B$ be an integral ring map. Then

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a) If $\phi^{-1}(\mathfrak{b}) \subseteq \mathfrak{a}$ then the induced ring map $A/\mathfrak{a} \to B/\mathfrak{b}$ (see (3.3.56)) is integral.
b) If S is a multiplicatively closed subset of A and $T \coloneqq \phi(S)$, then the induced ring map $S^{-1}A \to T^{-1}B$ is also integral.

Proposition 3.22.16 (Maximal ideals under integral extension)
Let $\phi : A \to B$ be an integral ring map. Suppose $\mathfrak{q}$ lies above $\mathfrak{p}$. Then

$$\mathfrak{p} \text{ is maximal} \iff \mathfrak{q} \text{ is maximal}$$

Proof. The map $\phi : A \to B$ induces an injective map $A/\mathfrak{p} \hookrightarrow B/\mathfrak{q}$ of integral domains by (3.3.56). By (3.22.15) this map is also integral. Note $A/\mathfrak{p}$ (resp. $B/\mathfrak{q}$) is a field if and only if $\mathfrak{p}$ (resp. $\mathfrak{q}$) is a maximal ideal by (3.3.59). Then we may apply (3.22.14) to show the equivalence. ∎

Lemma 3.22.17 (Non-zero Ideals meet the Base)
Let $A \subset R$ be integral domains such that every element of R is algebraic over $\mathrm{Frac}(A)$*. Then $\mathfrak{a} \cap A \neq 0$ for every non-zero ideal $\mathfrak{a} \lhd R$.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $0 \neq x \in \mathfrak{a}$. Clearing denominators, x satisfies a non-trivial relation $a_n x^n + ... + a_1 x + a_0 = 0$ with $a_i \in A$, and we take n minimal. Then $a_0 \neq 0$, for otherwise dividing by x (R is a domain) gives a non-trivial relation of smaller degree. Therefore

$$0 \neq a_0 = -x(a_n x^{n-1} + ... + a_1) \in xR \subseteq \mathfrak{a}$$

so $a_0 \in \mathfrak{a} \cap A$. ∎

Proposition 3.22.18 (Properties of integral extensions)
Let $\phi : A \to B$ be an integral ring map then it has

*a) the **Lying Over** property*
*b) the **Incomparability** property*
*c) the **Going Up** property*

Proof. For any prime ideal $\mathfrak{p} \lhd A$ we have the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_T} \\ A_{\mathfrak{p}} & \overset{\tilde{\phi}}{\dashrightarrow} & B_{\mathfrak{p}} \end{array}$$

where $S \coloneqq A \setminus \mathfrak{p}$ and $T \coloneqq \phi(S)$. By (3.5.5) there exists a morphism $\tilde{\phi}$, and by (3.22.15) it is integral. Define $\mathfrak{m} \coloneqq \mathfrak{p}A_{\mathfrak{p}}$ to be the unique maximal ideal of $A_{\mathfrak{p}}$.

a) As we assume $\ker(\phi) \subseteq \mathfrak{p}$ we know $B_{\mathfrak{p}} \neq 0$ (3.5.40). Let $\mathfrak{n}$ be a maximal (and hence prime) ideal of $B_{\mathfrak{p}}$. Then $\mathfrak{q} \coloneqq i_T^{-1}(\mathfrak{n})$ is a prime ideal of B such that $\mathfrak{q} \cap T = \emptyset$. In addition by (3.22.16) $\tilde{\phi}^{-1}(\mathfrak{n})$ is a maximal ideal, and therefore by uniqueness $\mathfrak{m} = \tilde{\phi}^{-1}(\mathfrak{n})$. By commutativity of the diagram we then have $\phi^{-1}(\mathfrak{q}) = \mathfrak{p}$ as required.

(Stacks) As an alternative argument to show existence of $\mathfrak{q}$ by (3.21.5) it's enough to show that $\mathfrak{p}B_{\mathfrak{p}}$ is proper (by assumption $\ker(\phi) \subseteq \mathfrak{p}$ so $B_{\mathfrak{p}} \neq 0$). By the diagram above $\mathfrak{p}B_{\mathfrak{p}} = \tilde{\phi}(\mathfrak{p}A_{\mathfrak{p}})B_{\mathfrak{p}}$. Therefore it's enough to consider the case $(A, \mathfrak{m})$ local and to show $\phi(\mathfrak{m})B$ is proper. Suppose $1 \in \phi(\mathfrak{m})B$ then

$$1 = \sum_{i=1}^{n} \phi(a_i) b_i \quad a_i \in \mathfrak{m}_A \; b_i \in B.$$

By (3.22.4) the subring $B' \coloneqq \phi(A)[b_1, ..., b_n] \subset B$ is a finite A-module. Furthermore $1 \in \mathfrak{m}B'$ whence $\mathfrak{m}B' = B'$ and by Nakayama's Lemma (3.19.5) $B' = 0$, a contradiction.

b) Suppose $\mathfrak{p} = \phi^{-1}(\mathfrak{q}) = \phi^{-1}(\mathfrak{q}')$ and $\mathfrak{q} \subseteq \mathfrak{q}'$. Let $\mathfrak{n} = \mathfrak{q}B_{\mathfrak{p}}$ and $\mathfrak{n}' = \mathfrak{q}'B_{\mathfrak{p}}$. Clearly $\mathfrak{n} \subseteq \mathfrak{n}'$. By commutativity of the diagram $i_S^{-1}(\tilde{\phi}^{-1}(\mathfrak{n})) = \phi^{-1}(\mathfrak{q}) = \mathfrak{p}$. By (3.5.18) extending the ideals to $A_{\mathfrak{p}}$ shows $\tilde{\phi}^{-1}(\mathfrak{n}) = \mathfrak{m}$, and similarly for $\mathfrak{n}'$. By (3.22.16) both $\mathfrak{n}$, $\mathfrak{n}'$ are maximal so $\mathfrak{n} = \mathfrak{n}'$. By (3.5.19) $\mathfrak{q} = \mathfrak{q}'$.

c) Suppose we have prime ideals $\mathfrak{p} \subsetneq \mathfrak{p}'$ and $\mathfrak{q}$ is a prime ideal lying above $\mathfrak{p}$. Consider the commutative diagram

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$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p} & \xrightarrow{\tilde{\phi}} & B/\mathfrak{q} \end{array}$$

The induced map $\tilde{\phi}$ is integral (3.22.16). By a) there is a prime ideal of $B/\mathfrak{q}$ lying above $\mathfrak{p}'/\mathfrak{p}$, which is of the form $\mathfrak{q}'/\mathfrak{q}$ for $\mathfrak{q} \subseteq \mathfrak{q}'$ prime (3.3.56). Then from the diagram we see $\phi^{-1}(\mathfrak{q}') = \mathfrak{p}'$ as required.

■

Proposition 3.22.19 (Coefficients of minimal polynomial)
Let $A \subseteq B$ be integral domains, A is integrally closed, and define $K = \mathrm{Frac}(A)$ and $L = \mathrm{Frac}(B)$. For $b \in B$ integral over $\mathfrak{a} \lhd A$ we have the non-leading coefficients of $m_b(X)$ are integral over $\mathfrak{a}$ and therefore lie in $\sqrt{\mathfrak{a}}$.

Note if b is only assumed to be integral over A then the coefficients of $m_b(X)$ lie in A.

Both $\mathrm{Norm}_{L/K}(b)$ and $\mathrm{Tr}_{L/K}(b)$ lie in A.

Proof. Let M/K be a normal closure for L/K (...). By (3.22.11) the integral closure of $\mathfrak{a}$ in M is simply $\sqrt{\mathfrak{a}}$. Then the minimal polynomial $m_b(X)$ splits completely in M and by (3.17.68) all the roots b_i are conjugate by $\mathrm{Aut}(M/K)$. In particular it's clear that b_i are integral over $\mathfrak{a}$, and so lie in $\sqrt{\mathfrak{a}}$. The coefficients of $m_b(X)$ are polynomials in the b_i, and so by the observation above are also lie in $\sqrt{\mathfrak{a}}$ (and are integral over $\mathfrak{a}$).

We may consider the case $\mathfrak{a} = A$ to deduce that $m_b(X)$ is a monic polynomial with coefficients in A. By (3.17.131) then the coefficients of $\mathrm{Pc}_{L/K}(b; X)$ lie in A, and so by (3.17.126) so do the norm and trace. ■

Lemma 3.22.20 (Colon Ideals of the Radical of an Extended Prime)
Let $A \subseteq B$ be an integral ring extension of domains with A integrally closed, and $\mathfrak{p} \lhd A$ a prime ideal. Then for every $x \in A \setminus \mathfrak{p}$ the colon ideal $(\sqrt{\mathfrak{p}B} : x) := \{s \in B \mid xs \in \sqrt{\mathfrak{p}B}\}$ equals $\sqrt{\mathfrak{p}B}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The inclusion $\sqrt{\mathfrak{p}B} \subseteq (\sqrt{\mathfrak{p}B} : x)$ is obvious as $\sqrt{\mathfrak{p}B}$ is an ideal, so it remains to show $(\sqrt{\mathfrak{p}B} : x) \subseteq \sqrt{\mathfrak{p}B}$. Let $s \in B$ with $y := xs \in \sqrt{\mathfrak{p}B}$. As $x \neq 0$ we may assume $y \neq 0$, for otherwise $s = 0$. By (3.22.11) y is integral over $\mathfrak{p}$, so by (3.22.19) the minimal polynomial of y over $K := \mathrm{Frac}(A)$ is

$$m_{y,K}(X) = X^r + u_1 X^{r-1} + ... + u_r \quad u_i \in \sqrt{\mathfrak{p}} = \mathfrak{p}$$

Since $x \in K^\times$, the element $s = yx^{-1}$ has minimal polynomial over K obtained by scaling the roots of $m_{y,K}$ by x^{-1}, which preserves being monic, irreducible and of degree r:

$$m_{s,K}(X) = X^r + v_1 X^{r-1} + ... + v_r \quad v_i := \frac{u_i}{x^i}$$

As $s \in B$ is integral over A and A is integrally closed, the coefficients of $m_{s,K}$ lie in A by (3.22.19), so $v_i \in A$. Now $v_i x^i = u_i \in \mathfrak{p}$ with $x \notin \mathfrak{p}$ and $\mathfrak{p}$ prime, so $v_i \in \mathfrak{p}$ for all i. Therefore s is integral over $\mathfrak{p}$ and $s \in \sqrt{\mathfrak{p}B}$ by (3.22.11). ■

Proposition 3.22.21 (Going Down)
Let $A \subseteq B$ be integral ring extension such that A is an integrally closed domain and B is an integral domain. Then it has the Going Down property.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\mathfrak{p} \subsetneq \mathfrak{p}'$ be prime ideals of A and $\mathfrak{q}'$ a prime ideal lying over $\mathfrak{p}'$. We wish to find a prime ideal $\mathfrak{q} \subseteq \mathfrak{q}'$ lying over $\mathfrak{p}$ (clearly inclusion must be strict).

Consider the inclusion of rings $A \subseteq B_{\mathfrak{q}'}$. Then by (3.21.5) $\mathfrak{p}$ lies under a prime of $B_{\mathfrak{q}'}$ if and only if $\mathfrak{p} = \mathfrak{p}^{ec} = \mathfrak{p}B_{\mathfrak{q}'} \cap A$. If this is the case then it is of the form $\mathfrak{q}B_{\mathfrak{q}'}$ for some prime ideal $\mathfrak{q} \subseteq \mathfrak{q}'$ (3.5.42) of B. It's clear that $\mathfrak{q}$ lies over $\mathfrak{p}$.

Note in general that $\mathfrak{p} \subseteq \mathfrak{p}^{ec}$, so we only need to demonstrate the reverse inclusion. Choose $x \in \mathfrak{p}B_{\mathfrak{q}'} \cap A$ and suppose for contradiction that $x \notin \mathfrak{p}$. By (3.5.18) $\mathfrak{p}B_{\mathfrak{q}'} = S^{-1}(\mathfrak{p}B)$ where $S = B \setminus \mathfrak{q}'$, so $x = \frac{y}{s}$ for $y \in \mathfrak{p}B$ and $s \in B \setminus \mathfrak{q}'$. Then $xs = y \in \mathfrak{p}B \subseteq \sqrt{\mathfrak{p}B}$ with $x \in A \setminus \mathfrak{p}$, so $s \in (\sqrt{\mathfrak{p}B} : x) = \sqrt{\mathfrak{p}B}$ by (3.22.20). But $\mathfrak{p}B \subseteq \mathfrak{p}'B \subseteq \mathfrak{q}'$ as $\mathfrak{q}'$ lies over $\mathfrak{p}' \supseteq \mathfrak{p}$, and $\mathfrak{q}'$ is prime, so $\sqrt{\mathfrak{p}B} \subseteq \mathfrak{q}'$ and $s \in \mathfrak{q}'$, which contradicts the choice of s. Therefore $x \in \mathfrak{p}$ as required. ■

Proposition 3.22.22
Let B, C be commutative A-algebras such that B is integral over A. Then $C \otimes_A B$ is integral over C.

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Proof. Let $\overline{C}$ be the subring of $C \otimes_A B$ of elements which are integral over C (3.22.7). Consider an elementary tensor $c \otimes b$ then by definition there is $P \in A[X]$ such that $P(b) = 0$. Then

$$P(c \otimes b) = \sum_{i=0}^{n} a_i (c \otimes b)^i = \sum_{i=0}^{n} c^i \otimes a_i b^i = c^i \otimes \sum_i a_i b^i = c^i \otimes P(b) = 0$$

This shows that $\overline{C}$ contains the elementary tensors and therefore is equal to $C \otimes_A B$. ∎

Lemma 3.22.23
Let $\phi : A \to B$ be a finite ring homomorphism. Suppose $B = (f_1, ..., f_n)$ and the composite map $\phi_i : A \to B_{f_i}$ is finite for all $i = 1...n$, then so is ϕ.

Proof. There are finitely many elements $\left\{ \frac{b_{ij}}{f_i^{n_{ij}}} \right\}_j$ which generate B_{f_i} as an A-module. Let B' be the A-submodule of B generated by all the b_{ij}. Then $B'_{f_i} = B_{f_i}$ as B-modules. By (3.5.47) we have $B' = B$ and therefore B is finitely generated as an A-module. ∎

Proposition 3.22.24 (Krull Dimension is preserved under integral maps)
Let $\phi : A \to B$ be a ring map.

a) Going Up $\implies \dim B \geq \dim(A/\ker(\phi))$
b) Incomparability $\implies \dim B \leq \dim(A/\ker(\phi))$

In particular ϕ integral and injective implies $\dim A = \dim B$.

Proof. Without loss of generality we can assume that ϕ is injective. The two cases follow by lifting chains of prime ideals from A (resp. B) to B (resp. A), and checking that distinct is maintained.

The final statement follows from (3.22.18). ∎

Corollary 3.22.25
Let $\phi : A \to B$ be integral and $\mathfrak{b} \lhd B$, then $\dim \phi^{-1}(\mathfrak{b}) = \dim(\mathfrak{b})$.

3.22.1 Integral Closure in Finite Extensions

Proposition 3.22.26
Let L/K be an algebraic field extension and A a subring with quotient field K. Then the integral closure $B := \overline{A}^L$

a) has quotient field L,
b) spans L as a K-vector space and
c) is integrally closed.

Furthermore $\dim B = \dim A$.

Proof. We may show a),b) if we prove $L = (A^\star)^{-1} B$. Consider $x \in L$, then by definition it is integral over $K = (A^\star)^{-1} A$ so by (3.22.8) $ax \in B$, and therefore $x \in (A^\star)^{-1} B$ as required.

It is integrally closed by (3.22.7) and $\dim B = \dim A$ from (3.22.24). ∎

Proposition 3.22.27
Let L/K be a finite field extension, A a subring with quotient field K and $B = \overline{A}^L$. Suppose that A is a principal ideal domain and B is a finite A-module. Then B is a finite free A-module of rank $[L : K]$.

Proof. By (3.14.7) B is a finite free A-module. Let $b_1, ..., b_r$ be an A-basis, then by (3.22.26) this spans L as a K-vector space. Similarly by clearing denominators the given set is K-linearly independent, so it is a K-basis of L and $r = [L : K]$ (3.3.126) as required. ∎

Proposition 3.22.28 (Finite Separable Extension of Normal Domain)
Let L/K be a finite separable extension, with A an integrally closed Noetherian subring with quotient field K. Then the integral closure $\overline{A}^L$ is a finite A-module.

In particular $\overline{A}^L$ is Noetherian and $\dim A = \dim \overline{A}^L$.

If in addition A is a principal ideal domain, then $\overline{A}^L$ is a free A-module of rank $[L : K]$.

Proof. By (3.17.134) the trace map $L \times L \to K$ is a perfect pairing. By (3.22.26) we may choose $x_1, ..., x_n \in B$ a K-basis for L where $n = [L : K]$. Denote $x_1^\star, ..., x_n^\star \in L$ by the (unique) basis dual under this pairing satisfying

$$\mathrm{Tr}_{L/K}(x_i x_j^\star) = \delta_{ij}$$

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We claim that $B := \overline{A}^L$ satisfies

$$B \subset Ax_1^\star + ... + Ax_n^\star.$$

For every $b \in B$ we have $b = \sum_{i=1}^n \lambda_i x_i^\star$ and some $\lambda_i \in K$. Then by linearity $\mathrm{Tr}_{L/K}(bx_i) = \lambda_i$. By (3.22.19) we have $\lambda_i \in A$. The right hand side is a finitely generated A-module, which is therefore Noetherian (3.13.4). B is a submodule and therefore also finitely generated as an A-module (3.13.3).

Then by (3.13.4) B is Noetherian as an A-module, therefore a-fortiori Noetherian as a B-module (or equivalently as a ring).

If A is a principal ideal domain then $\overline{A}^L$ is a finite free A-module of rank $[L : K]$ by (3.22.27). ■

Lemma 3.22.29
Let $A \subset B \subset C$ be subrings. Then $\overline{A}^C \cap B = \overline{A}^B$.

Lemma 3.22.30
Let $A \subset B \subset C$ be rings such that B is integral over A. Then $\overline{A}^C = \overline{B}^C$.

Lemma 3.22.31
Let A be a Noetherian integral domain with quotient field K and L/K be a finite extension.

Suppose that L'/L is a finite extension such $\overline{A}^{L'}$ is a finite A-module, then so is $\overline{A}^L$.

Proof. By (3.13.4) $\overline{A}^{L'}$ is Noetherian as an A-module, whence by (3.22.29) $\overline{A}^L \subset \overline{A}^{L'}$ is a finitely-generated A-module. ■

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